

Relative anchor, endpoint-first anisotropy, and canonical covariance

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1. Purpose and scope

The objective of this note is not to manufacture a preferred spatial phase. It asks what follows from the registered finite-cutoff action and covariance, independently of whether a later physical minimizer is BCC, another modulated state, uniform, disordered, or absent. No phase-selection statement is used or obtained.

R-145 leaves three apparent burdens on the direct T-050 trace-excess route: an absolute chart anchor, temporal transport of the anisotropic terminal covariance, and the scalar-principal signed owner. Two can be removed without changing the action.

First, A7 fixes the zero-control additive normalization and R-104 identifies the zero-control terminal law. Therefore the R-145 identity is exactly relative and no separate absolute chart-anchor estimate is needed. Second, one must telescope the complete owner before splitting the endpoint covariance. The anisotropic term is then bounded by

$$\frac{1}{100}Y_6(h) + 24 \tag{1.1}$$

with zero source loss. This is not a blockwise temporal transfer.

Third, R-087 permits a distinct fixed-cutoff Boué–Dupuis problem built from the constant Brownian factor of the same endpoint covariance. Its temporal covariance increments are proportional and hence admit an exact positive scalar-principal/anisotropic split. The variational value is unchanged, but the old chart’s visit, low, trace, and forest owners are not transported. They must be reconstructed in the new chart.

Finally, strict-past scalar common-noise rows have an exact nonpositive conditional defect. A bounded same-root coefficient supplies an exact positive counterfixture, so canonical covariance alone does not close the scalar sea. The remaining analytic target is the complete contemporaneous scalar signed relative owner with future variance, forest, and returned low retained once.

Thus R-146 is a genuine reduction, not T-050 closure. A13, Nelson, removal, the interacting measure, every phase question, and Sector A remain open.

2. Zero-control relative-anchor theorem

Fix a cutoff J and a temporally faithful chart π . R-145 gives the complete identity

$$\mathbb{E}V_J^{\text{en}}(Z_h) = a_{J,\pi} - \mathcal{T}_{J,\pi}(h), \quad \mathcal{T}_{J,\pi}(h) = \frac{1}{2}\{\|\mathcal{U}_\pi(h)\|^2 - \|\Phi_\pi(h)\|^2\}, \tag{2.1}$$

where $a_{J,\pi}$ is independent of h . R-104 supplies

$$Z_\pi(0) \stackrel{\text{law}}{=} X_J, \tag{2.2}$$

and the A7 covariance-normal definition supplies the external centering

$$\mathbb{E}V_J^{\text{ren}}(X_J) = 0. \quad (2.3)$$

Theorem 2.1 (exact relative anchor). Under (2.1)–(2.3),

$$\boxed{a_{J,\pi} = \mathcal{T}_{J,\pi}(0), \quad \mathbb{E}V_J^{\text{ren}}(Z_h) = -[\mathcal{T}_{J,\pi}(h) - \mathcal{T}_{J,\pi}(0)].} \quad (2.4)$$

Proof. Set $h = 0$ in (2.1), use (2.2) to replace the terminal law by X_J , and then use (2.3). This gives $0 = a_{J,\pi} - \mathcal{T}_{J,\pi}(0)$. Substitution into (2.1) proves (2.4). $\square$

Consequently a uniform relative estimate

$$\mathcal{T}_{J,\pi}(h) - \mathcal{T}_{J,\pi}(0) \leq \eta X(h) + \zeta Y_6(h) + B, \quad (2.5)$$

with

$$\eta < \frac{5}{11}, \quad \zeta < \frac{27}{100}, \quad (2.6)$$

implies the direct T-050 lower bound with constant B . No additional assumption $a_{J,\pi} \geq -C_0$ is required on this relative route.

This does not refute NG-2026-07-30-A13-NORMALIZED-GIBBS-DOOB-ABSOLUTE-ANCHOR. Normalized laws are invariant under $V \mapsto V + c$ and cannot discover the constant c . Here the external A7 covariance-normal convention has already fixed that gauge through (2.3). Also, (2.4) does not prove that $\mathcal{T}_{J,\pi}(0)$ or $a_{J,\pi}$ is separately bounded uniformly in cutoff or chart; the subtraction occurs before the limiting estimate.

3. Complete endpoint-first anisotropic payment

Let the registered terminal covariance split be

$$\Gamma_J = \Gamma_{0,J} + \Gamma_{R,J}, \quad \Gamma_{R,J} \succeq 0. \quad (3.1)$$

The point is to apply (3.1) only after the R-070/R-104 complete signed owner has telescoped to the endpoint. Let P be the registered R-145 orthogonal output projection. For an endpoint W , define

$$T_{\text{an}}(W) = \frac{1}{2} \mathbb{E} \int \text{Tr}\{PC_6(W)\Gamma_{R,J}C_6(W)^*P\} dx \geq 0. \quad (3.2)$$

Linearity of the terminal covariance trace gives

$$\mathcal{T}_{J,\pi}(h) = \mathcal{T}_{J,\pi}^{(0)}(h) + T_{\text{an}}(Z_h). \quad (3.3)$$

Here $\mathcal{T}^{(0)}$ is the full endpoint residual with the scalar-principal covariance; it is not a separately proved scalar law. Equation (3.3) is made after complete recombination and asserts no positive anisotropic allocation to an individual temporal visit.

Theorem 3.1 (endpoint-first anisotropic reduction). At regulator bound one, the R-145 terminal theorem and (3.2)–(3.3) imply

$$\boxed{\Delta \mathcal{T}_{J,\pi} \leq \Delta \mathcal{T}_{J,\pi}^{(0)} + \frac{1}{100} Y_6(h) + 24, \quad \Delta \mathcal{T}(h) = \mathcal{T}(h) - \mathcal{T}(0).} \quad (3.4)$$

The added source coefficient is exactly zero.

Proof. Subtract (3.3) at zero control:

$$\Delta \mathcal{T} = \Delta \mathcal{T}^{(0)} + T_{\text{an}}(Z_h) - T_{\text{an}}(Z_0) \leq \Delta \mathcal{T}^{(0)} + T_{\text{an}}(Z_h), \quad (3.5)$$

because $T_{\text{an}}(Z_0) \geq 0$. R-145 gives

$$T_{\text{an}}(Z_h) \leq \frac{1}{100} Y_6(h) + 24. \quad (3.6)$$

Combining (3.5) and (3.6) proves (3.4). $\square$

The scalar-principal remainder therefore retains the exact windows

$$\boxed{\eta_{\text{sc}} < \frac{5}{11}, \quad \zeta_{\text{sc}} < \frac{27}{100} - \frac{1}{100} = \frac{13}{50}.} \quad (3.7)$$

No source headroom is spent on anisotropy, and the constant 24 is paid once at the complete endpoint.

4. Arbitrary blockwise anisotropic transport is unavailable

R-129 already shows that total Fourier diagonality does not make arbitrary temporal increments commute with a shell projector. A stronger obstruction shows that even a positive suballocation of a fixed anisotropic remainder need not exist.

Put

$$P_+ = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad P_- = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad P_+ + P_- = I_2, \quad (4.1)$$

and request a decomposition of

$$R = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad (4.2)$$

as $R = R_+ + R_-$ with $0 \preceq R_{\pm} \preceq P_{\pm}$. The elementary dominance lemma is: if $0 \preceq A \preceq P$ and P is an orthogonal projection, then for $x \in \text{Ker } P$, $\langle x, Ax \rangle = 0$; hence $A^{1/2}x = 0$, so $\text{Ran } A \subseteq \text{Ran } P$. Since $P_{\pm}$ have rank one, this support inclusion forces

$$R_{\pm} = a_{\pm} P_{\pm}, \quad 0 \leq a_{\pm} \leq 1. \quad (4.3)$$

But both diagonal entries of $a_+ P_+ + a_- P_-$ equal $(a_+ + a_-)/2$, whereas the diagonal entries of R are one and zero. No such decomposition exists.

This proves NG-2026-08-02-A13-ARBITRARY-TEMPORAL-ANISOTROPIC-POSITIVE-SUBALLOCATION. It refutes the method, not the full action bound. The legal repair is Theorem 3.1: telescope first, split once, and use the favourable zero-control term.

5. Canonical covariance as an exact alternative chart

Fix J and write an arbitrary endpoint factorization as

$$X_J = \int_0^T K_J(t) dB_t, \quad C_J = \int_0^T K_J(t) K_J(t)^* dt. \quad (5.1)$$

Define on a fresh canonical Brownian space

$$K_{\text{can}}(t) = T^{-1/2} C_J^{1/2}, \quad \tilde{X}_J = \int_0^T K_{\text{can}}(t) d\tilde{B}_t. \quad (5.2)$$

Then

$$\tilde{X}_J \stackrel{\text{law}}{=} X_J \sim N(0, C_J). \quad (5.3)$$

Theorem 5.1 (fixed-cutoff canonical covariance normal form). Let G_J obey the fixed-cutoff continuity, lower-bound, and local polynomial-envelope hypotheses of R-087 Theorem 8.1. For every $q > 0$,

$$-\frac{1}{q} \log \mathbb{E} e^{-q G_J(X_J)} = \inf_{v \in \mathcal{A}_{cs}} \mathbb{E} \left[G_J \left(\tilde{X}_J + \frac{C_J^{1/2}}{\sqrt{T}} \int_0^T v_t dt \right) + \frac{1}{2q} \int_0^T \|v_t\|^2 dt \right]. \quad (5.4)$$

At the Nelson exponent $q = 10/9$, the cost coefficient is $9/20$.

Proof. Equation (5.3) makes the left side equal to the original endpoint value. Apply R-087 Theorem 8.1 to the deterministic factor K_{can} . Its restriction to bounded smooth cylindrical-simple controls is exact. This proves (5.4). $\square$

For a deterministic partition $I_b = (t_{b-1}, t_b]$, put $\tau_b = |I_b|/T$. The field increment covariance is exactly

$$\Delta C_b = \tau_b C_J. \quad (5.5)$$

If $C_J = C_{0,J} + R_J$ with $C_{0,J}, R_J \succeq 0$, an expanded Brownian space gives the rectangular factor

$$K_{\text{rect}} = T^{-1/2} [C_{0,J}^{1/2} \quad R_J^{1/2}], \quad (5.6)$$

and hence the exact positive split

$$\Delta C_b = \tau_b C_{0,J} + \tau_b R_J. \quad (5.7)$$

No false identity $C_J^{1/2} = C_{0,J}^{1/2} + R_J^{1/2}$ is used. In the A1 translation-invariant cutoff, these covariance operators and Fourier shell projections are simultaneous multipliers, so

$$[\Delta C_b, \Pi_m] = 0, \quad \Delta C_{b,m} = \tau_b \Pi_m C_J \Pi_m. \quad (5.8)$$

For a blockwise constant control $v_t = u_b$ on I_b , put $h_b = \sqrt{\tau_b T} u_b$. Then

$$\int_0^T \|v_t\|^2 dt = \sum_b \|h_b\|^2, \quad \int_0^T K_{\text{can}} v_t dt = C_J^{1/2} \sum_b \sqrt{\tau_b} h_b. \quad (5.9)$$

Thus every tail S obeys

$$\left\| C_J^{1/2} \sum_{b \in S} \sqrt{\tau_b} h_b \right\|^2 \leq \|C_J\|_{\text{op}} \left(\sum_{b \in S} \tau_b \right) \sum_{b \in S} \|h_b\|^2. \quad (5.10)$$

The primary and independent certificates verify this normalization on decimal and rational fixtures. A source in $\text{Ker } C_J$ costs energy but produces no endpoint displacement, so it is zero at the infimum and may be removed by the range quotient.

Theorem 5.1 preserves the complete endpoint variational value. It does not identify a control from the old chart with one in the new chart, and does not preserve individual reveal, trace, forest, or returned-low terms. The original arbitrary-chart T-050 theorem is therefore not proved. A new uniform lower bound established directly in (5.4), however, would imply the fixed-cutoff Nelson estimate by the exact R-087 variational equality.

6. Scalar common-noise sign and its exact boundary

Let $\mathcal{F}_{\text{past}}$ be a strict-past sigma-field. Conditionally on it, let A and u be fixed, let $g \sim N(0, \Gamma)$, and let $\zeta \sim N(0, \text{I})$ be independent. Define

$$U = A\Gamma^{1/2}\zeta, \quad \Phi = A(g + u). \quad (6.1)$$

Lemma 6.1 (strict-past scalar common-noise sign).

$$\boxed{\mathbb{E}[\|U\|^2 - \|\Phi\|^2 \mid \mathcal{F}_{\text{past}}] = -\|Au\|^2 \leq 0.} \quad (6.2)$$

Proof. Both Gaussian squares have the same conditional covariance trace $\text{Tr}(A\Gamma A^*)$. The shifted current square has the additional term $\|Au\|^2$, and its mixed Gaussian term has conditional mean zero. $\square$

For two endpoints using the same Gaussian and probe, the corresponding polarized identity is

$$\mathbb{E}\langle U_q, U_p \rangle - \mathbb{E}\langle \Phi_q, \Phi_p \rangle = -\langle A_q u_q, A_p u_p \rangle. \quad (6.3)$$

Equation (6.3) is exact, but a mixed inner product has no sign by itself. Only a legally recombined diagonal or an established adjoint owner may use it as a sign statement.

The strict-past hypothesis is load-bearing. Let independent $g, \zeta \sim N(0, 1)$, and for $t > 0$ define the bounded smooth coefficient

$$A_t(g) = ge^{-tg^2/2}, \quad U = A_t(g)\zeta, \quad \Phi = A_t(g)g. \quad (6.4)$$

Gaussian differentiation gives

$$\mathbb{E}\|U\|^2 = (1 + 2t)^{-3/2}, \quad \mathbb{E}\|\Phi\|^2 = 3(1 + 2t)^{-5/2}, \quad (6.5)$$

and therefore

$$\boxed{\mathbb{E}(\|U\|^2 - \|\Phi\|^2) = \frac{2(t-1)}{(1+2t)^{5/2}} > 0 \quad (t > 1).} \quad (6.6)$$

Relative to the trace square, the defect is

$$\frac{2(t-1)}{1+2t} \longrightarrow 1. \quad (6.7)$$

At $t = 2$ this ratio is exactly $2/5$. This proves NG-2026-08-02-A13-CANONICAL-COVARIANCE-AUTOMATIC-SCALAR-CANCELLATION. It is a counterexample to the inference from covariance equality, not a counterexample to the complete production action.

Canonical proportional covariance also does not manufacture a common terminal martingale. With independent centered roots ξ_1, ξ_2 ,

$$\Phi_1 = \xi_1, \quad \Phi_2 = 2\xi_1 + \xi_2, \quad \mathbb{E}[\Phi_2 \mid \mathcal{F}_1] = 2\xi_1 \neq \Phi_1. \quad (6.8)$$

Thus the R-125 common-terminal hypothesis remains a separate theorem.

7. Exact remaining target and two honest routes

After Theorems 2.1 and 3.1, the old-chart direct route has one global target:

$$\Delta\mathcal{T}_{J,\pi}^{(0)}(h) \leq \eta_{\text{sc}}X(h) + \zeta_{\text{sc}}Y_6(h) + B, \quad \eta_{\text{sc}} < \frac{5}{11}, \quad \zeta_{\text{sc}} < \frac{13}{50}. \quad (7.1)$$

Here $\mathcal{T}^{(0)}$ retains the full endpoint field and all scalar-principal trace, current, future-variance, forest, and low effects; it is not a coefficient-frozen scalar model. If (7.1) holds, Theorems 2.1 and 3.1 give

$$\mathbb{E}V_J^{\text{ren}}(Z_h) \geq -\eta_{\text{sc}}X(h) - (\zeta_{\text{sc}} + 1/100)Y_6(h) - (B + 24), \quad (7.2)$$

which is exactly inside the R-145 T-050 window.

The complete low cannot be estimated separately on the full progressive revisit class: the registered revisit fixture has quartic separated-low loss, quadratic source cost, and fixed terminal sextic. Cancellation must remain inside the signed relative owner.

The canonical alternative begins from Theorem 5.1 and follows this fail-closed sequence:

1. construct one complete terminal current in the canonical chart;
2. define candidate Doob rows by conditional expectation of that terminal;
3. verify the tower identity root by root and reject the route if (6.8) or any owner-incidence test fails;
4. rederive the R-063 forest, R-125 future variance, trace partition, and returned low directly in this chart, with no import of individual old owners;
5. remove the strict-past scalar diagonal by Lemma 6.1;
6. retain the contemporaneous coefficient, balanced cross, future variance, forest, and low in one signed residual and prove its lower bound;
7. invoke R-087 only after the complete lower bound is uniform in cutoff.

Schematically, the final scalar obligation has the form

$$\mathfrak{R}_{\text{sc}} = \mathbb{E}_f \int \{2\Re\langle U_0, \delta U \rangle + \|\delta U\|^2\} dt - \Delta \text{Forest}_{063}^{\text{sc}} + L_{\text{future/mean/low}}, \quad (7.3)$$

followed by

$$\mathfrak{R}_{\text{sc}} \leq \eta_{\text{sc}} X + \zeta_{\text{sc}} Y_6 + B. \quad (7.4)$$

Equation (7.3) names the owner-complete reconstruction target; it is not asserted as a proved production identity in the new chart.

A separate canonical-prefix estimate would have diagnostic source coefficient

$$4\beta_{\text{op}gR} = \frac{882195469063103}{8467200000000000} = 0.1041897521 \dots \quad (7.5)$$

and a Young integer ceiling 67 at sextic allocation 1/100. It is not a registered theorem because no exact production stage-owner tower has been proved. The production result is the endpoint-first coefficient zero and constant 24 in Theorem 3.1.

8. Proof and failure map

Object or route	Verdict	Evidence and next condition
Separate absolute chart anchor	Removed on direct relative route	A7 centering plus R-104 zero-control law gives Theorem 2.1; no separate uniform bound on $a_{J,\pi}$ is claimed.
Endpoint anisotropic owner	Proved and paid	Telescope complete owner first; use positivity at zero and R-145 once; cost $0 \cdot X + (1/100)Y_6 + 24$.
Arbitrary temporal positive anisotropic split	Failed method	The rank-one $P_{\pm}$ fixture cannot allocate $R = \text{diag}(1, 0)$.
Canonical endpoint covariance chart	Exact alternative	Same Gaussian endpoint law and R-087 CORE; increments are $\tau_b C_J$. Individual old owners are not preserved.
Strict-past common-noise	scalar diagonal	Proved favourable Lemma 6.1 gives $-\ Au\ ^2$.
Automatic scalar cancellation	same-root	Failed inference The bounded coefficient (6.4) has positive defect approaching the full trace.
Common-terminal family	Doob	Open Must be constructed and tower-tested; rootwise adaptation is insufficient.
Complete scalar forest/future/low residual	for-	Open, sole direct burden Prove (7.1) in the old chart or reconstruct and bound (7.3)–(7.4) in the canonical chart.
Physical phase or BCC selection		Outside result No minimizer comparison, Hessian phase test, or reference-vacuum ranking is performed.
T-050, Nelson, Sector A	Open	Require the complete scalar signed bound and the registered downstream gates.

The table is one dependency map, not ten independent claims. Successful, failed, and open routes are retained together so later work does not repeat an illegal temporal split or an automatic scalar-cancellation claim.

9. Devil’s-advocate and code-discipline review

- Objection: Theorem 2.1 contradicts the absolute-anchor no-go. DISMISSED.** The no-go concerns normalized data without an external additive convention. A7 supplies that convention. No separate uniform zero-control trace bound is inferred.
- Objection: The endpoint-first proof secretly transfers the anisotropic covariance through time. DISMISSED.** All temporal owners are recombined first. Only the terminal covariance trace is split, and the zero-control term is discarded in the favourable direction.
- Objection: Every positive temporal increment admits a positive anisotropic subincrement. UPHELD against the shortcut.** Equations (4.1)–(4.3) give an exact rank-one counterexample.
- Objection: Equal endpoint covariance preserves every old owner. UPHELD against the shortcut.** It preserves the Gaussian endpoint law and variational value only. Filtration-dependent visit, forest, future-variance, and low terms must be rebuilt.
- Objection: Proportional increments imply a common-terminal Doob family. UPHELD against the shortcut.** Equation (6.8) separates covariance normalization from the tower property.

6. **Objection: Canonical covariance makes the complete scalar sea nonpositive. UPHOLD against the shortcut.** The strict-past lemma is valid, while (6.4)–(6.7) show a positive same-root defect. The forest/low recombination is indispensable.
7. **Objection: The diagnostic coefficient and ceiling in (7.5) are a production bound. DISMISSED.** They are explicitly non-production diagnostics. The registered endpoint theorem has source coefficient zero and integer constant 24.
8. **Objection: A factor $1/2$, realification multiplicity, or source normalization may have been lost. MITIGATED.** The endpoint definition retains the outer $1/2$; the primary and independent scripts separately derive the R-145 constants, $h_b = \sqrt{\tau_b T} u_b$, the rational tail inequality, and exact thresholds. Numbers are computed from A1/R-145 authorities, not pasted as proof outputs.
9. **Objection: The argument assumes invertible covariance. DISMISSED.** The positive square root exists for singular covariance. Kernel controls produce no shift, retain positive cost, and are removed at the infimum.
10. **Objection: This result favors BCC or any other phase. DISMISSED.** No phase ansatz or energy ranking enters any proof. The result is entirely phase-neutral.
11. **Objection: These reductions close T-050. DISMISSED.** The complete contemporaneous scalar signed owner and its future/forest/low coordinates remain open. T-050, Nelson, and Sector A therefore remain open.

Limit checks are explicit: zero anisotropic remainder makes Theorem 3.1 trivial; a one-block canonical partition gives $\Delta C = C_J$; a kernel source has zero displacement; $u = 0$ saturates Lemma 6.1; and the same-root counterfixture changes sign exactly at $t = 1$. The proof is finite-cutoff; no cutoff, floor, or volume limit is inferred.

10. Result footer

The primary and non-importing independent certificates separately reconstruct the exact constants and fixtures. The integrated verifier reruns both, pins SHA-256 authorities, checks public surfaces, and verifies the rendered PDF.

Dependencies: A1, A7, R-070, R-087, R-104, R-125, R-129, R-145.

Reproduction command:

```
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/
a13_classii_canonical_covariance_relative_anchor_anisotropic_temporal_reduction_verify.py
```

Result ID: A13-CLASSII-CANONICAL-COVARIANCE-RELATIVE-ANCHOR-ANISOTROPIC-TEMPORAL-REDUCTION; Ledger ID: R-146.

Precise statement: relative anchor; endpoint anisotropy $\leq (1/100)Y6+24$ with zero source; canonical endpoint-payoff covariance chart; strict-past scalar sign, but no automatic same-root cancellation.

Scope: fixed cutoff/A1/A7/R-087; endpoint-only split; no old-owner transport.

Evidence grade: ANALYTIC, EXACT, EXECUTED, AUDITED.

Expected output: integrated PASS; primary 88/88; independent 95/95.

Falsification gate: any hash, sign, factor, covariance/cost, Gaussian, PDF, scope, public-surface, or release check fails.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: no old-owner transfer, complete scalar/low/forest/future owner, phase selection, T-050, A13, Nelson, measure, or Sector A.

Proof complete: false; T-050 closed: false; Sector A closed: false.

Next required action: tower-test the canonical owner and bound its scalar residual inside source 5/11 and sextic 13/50, or abandon this chart route.